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PyRUA-DataFarm

LLM Agent as a Cost-Efficient Teleoperator for Real-Ready Robot Data

Coding Agents as Teleoperators: Low-Cost, Real-Ready Robot Data

In one sentence Given one real calibration video and a one-sentence request, the agent rebuilds the task, expert and hidden checker in simulation; a video model renders multi-view video from real frames, followed by independent review. Training π0.5 on 100 such episodes yields 93/100 real-robot successes (84 with equal teleoperation), 87 vs 46 under new appearances, at ~$26 one-time per task and $0.80 per episode thereafter.

93 vs 84

Real-robot successes, original scene /100
Farm data vs equal-size teleoperation

87 vs 46

Unseen tablecloth + colored lighting /100
Farm data vs equal-size teleoperation

$21 + $0.80

One-time API cost per task (mean)
+ marginal cost per episode thereafter

Teleoperation is faithful but costly; synthetic data adds two costs

Teleop: faithful, pay per demo

- Every action runs on the real robot; frames show the deployment scene
- Each episode needs a robot + operator + resets after failures
- Cost grows linearly with data; a new task / scene / view means starting over

Cost 1: Human engineering

- MimicGen and DemoGen augment demos; in RoboTwin 2.0 an LLM writes tasks
- Yet each new task's scene, expert and success check still need manual setup and debugging
- The model writing a task often writes its own checks too

Cost 2: Visual gap

- Real2Sim2Real relies on reconstructed assets: cloth, lit screens, etc. are hard to rebuild and still look synthetic
- Video generation models (e.g., DreamGen) synthesize their own motion, giving no precise action labels
- New scenes require recollection

Goal: real-ready data Actions precise enough to train on directly, visuals real enough to deploy on the real robot directly, with no real demonstrations, and cheap for every new task.

Sim sets actions, real frames set looks, independent review admits

① Survey One calibration video + one request

② Model Agent rebuilds task; expert fit to hidden checker

③ Grow looks Video model: 3 views from real frames

④ Review, harvest Checked vs. sim state; only passes stored

Compiler-style lowering; each stage is approved by a non-producer

Grounding G

Cameras, scale, objects, joints; each quantity tagged measured / fitted / assumed / unknown; mass, friction etc. randomized, not guessed

Simulator qualification

Skill probes pick the passing variant with the fewest changes and issue a certificate (only a cloth probe so far)

Task contract T + expert C

Checker written & frozen by another role; author sees only dev-seed verdicts, not the checker or held-out seeds, blocking reward hacking

Backtrack + ledger

Defects return to the earliest responsible stage; all failed attempts are kept, so datasets carry full provenance

No self-grading: each fact has exactly one certifying source

	Camera	Appearance	Objects	Actions	State	Admission
Coding agent	○	○	○	○	○	✗
Calibration video	●	●	○	—	—	—
Certified simulator	—	✗	●	●	●	—
Renderer	—	○	✗	✗	✗	—
Real reference	—	●	✗	✗	✗	—
Verifier / reviewer	—	—	—	—	—	●

● **Can certify** ○ Propose only ✗ Cannot modify or certify — Not involved

The agent can propose anything but certify nothing; the renderer may change only appearance; references supply appearance, not position or motion.

Hidden checker + held-out seeds

Expert authors cannot see checker internals; admission is judged only on unseen initial states

Mutation testing calibrates verifiers

Mutate accepted clips (delete / duplicate / enlarge objects, shift events, swap views, recolor); its check must catch each corruption

Independent per-episode review

A separate agent inspects decoded videos against real reference frames and sim state, logging evidence

Role separation

Orchestrator, task designer, task verifier, expert author, render reviewer, process improver, each with explicit "must-nots"

Reskin, don't rewrite the scene: four general rendering designs

1

Sim structure, real looks

Edge maps of sim video drive ControlNet, locking arm pose, object positions and timing; real frames + real robot photos supply appearance

2

All cameras generated at once

Head and two wrist views tiled on one canvas and generated as one video, keeping lighting, materials and held objects consistent

3

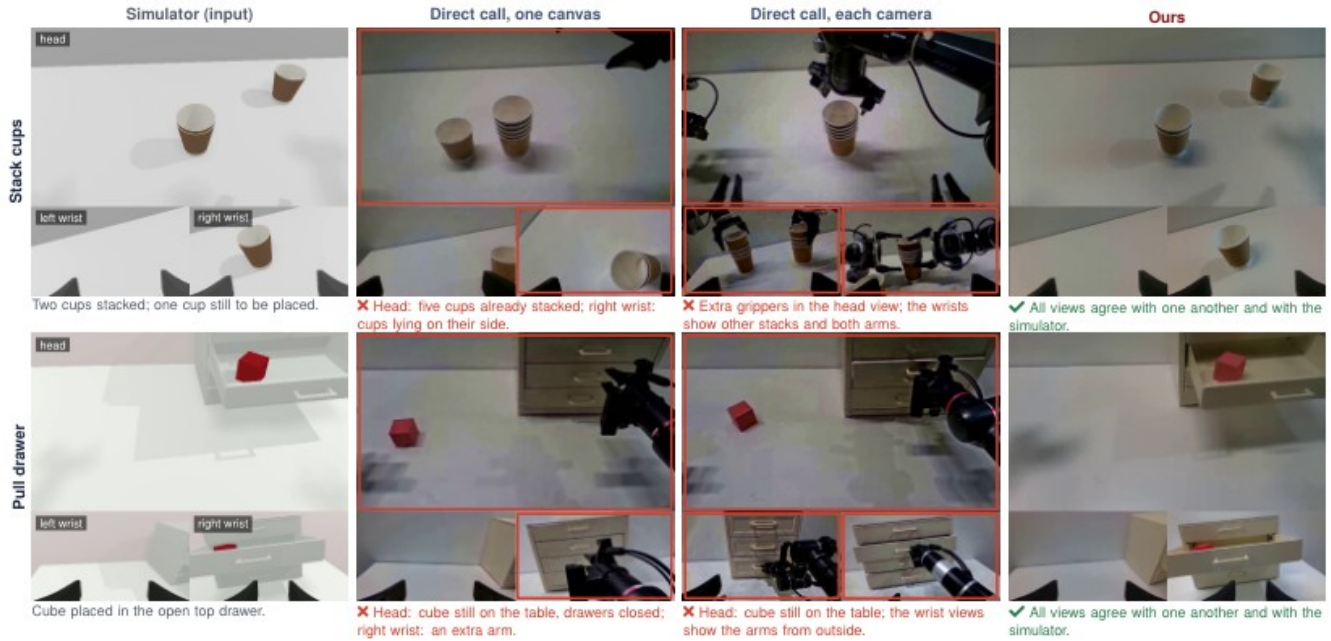
Refs/prompts add nothing

Reference frames match the sim task state; each prompt describes only its own chunk, avoiding copied layouts or premature future events

4

No drift in long clips

Generated in independent chunks, stitched where neighbors differ least; a VLM compares against sim video and flagged chunks are re-rendered



Direct video-model call: copies the reference frame's task state (e.g., "five cups already stacked"); per-camera generation hallucinates extra grippers / cup stacks.

Our renderer: the three views agree with each other and with the simulation.

Source: Fig. 8, § 4.5 (default renderer MiniMax-H3 / Sol-H3; Seedance also used)

5 real bimanual tasks; 3 training sets differ only in data source

Task	Challenge	Simulator	Success condition
Stack cups	Nesting thin rigid cups	SAPIEN	Three cups in one stack
Weigh apple	Scale with live display	SAPIEN	Apple resting on scale pan
Beat drum	Rhythmic contact timing	SAPIEN	Drum hit at least twice
Drawer	Articulated; pick-place-close	SAPIEN	Block inside, drawer closed
Fold towel	Deformable cloth	Isaac Sim	Towel folded twice

- Bimanual PiPER; per task only 1 calibration video (head + two wrist cameras) + a request for 100 episodes
- Rigid / articulated objects in SAPIEN (RoboTwin 2.0); towel as Isaac Sim particle cloth
- Operator agent: GPT-6 Astra, GPT-5.6-Sol or Claude Opus 5.5

π0.5 post-training, N = 100, same recipe

Real teleop

100 teleoperated episodes in the original scene

Sim + DR

Same sim trajectories, sim-rendered + domain randomization; isolates "how frames are made"

DataFarm

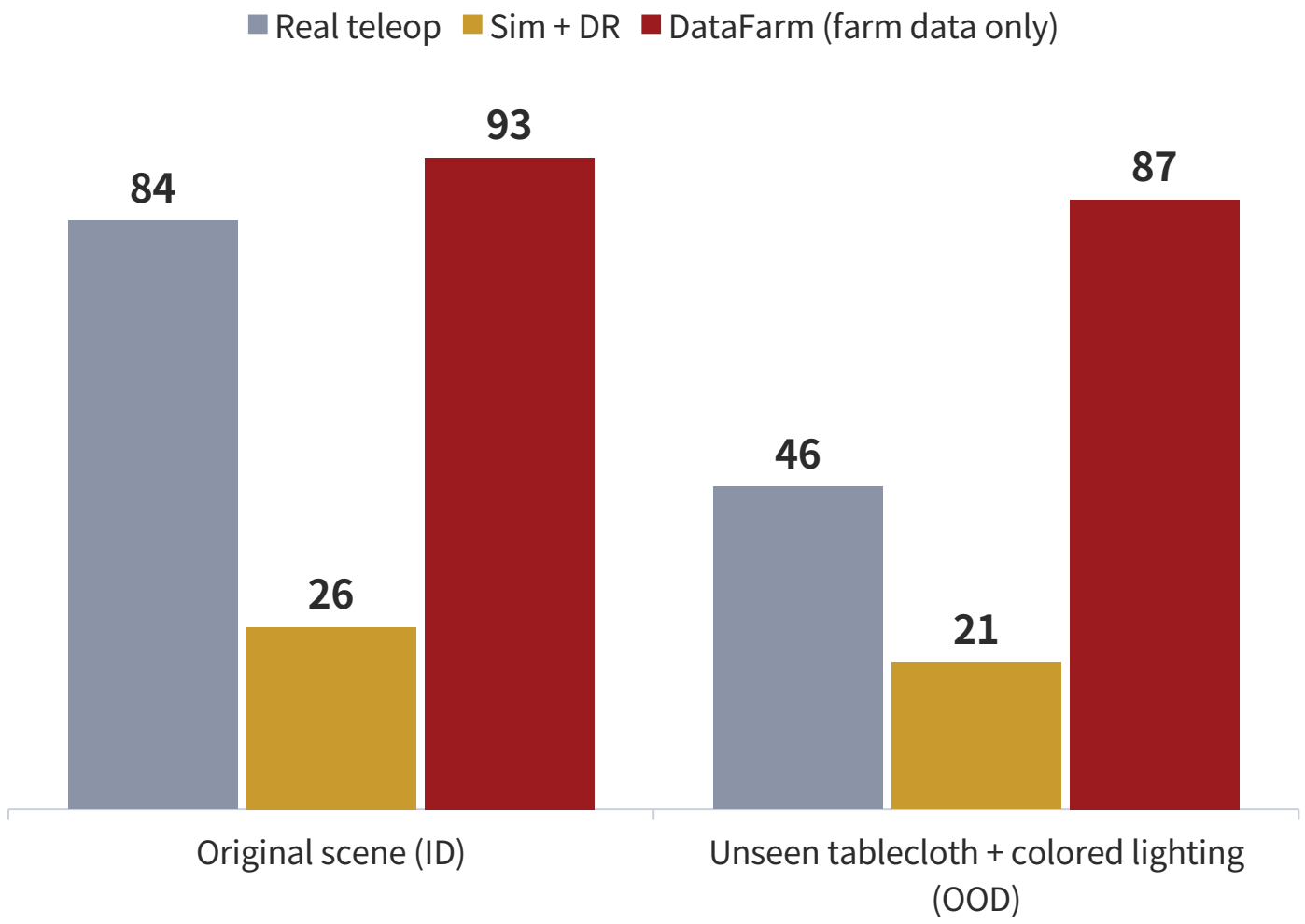
Farm-rendered data only; includes text-specified tabletop materials and lighting changes

Real-robot eval: 20 trials per task per condition

ID: original scene

OOD: tablecloths never seen in training (dark-gray felt / green plaid), with room light color changed during trials

Farm data alone: ties teleop in-scene, leads widely under new looks



93 vs 84

ID: ties or beats teleop on 4 of 5 tasks; cup stacking 1 success fewer

87 vs 46

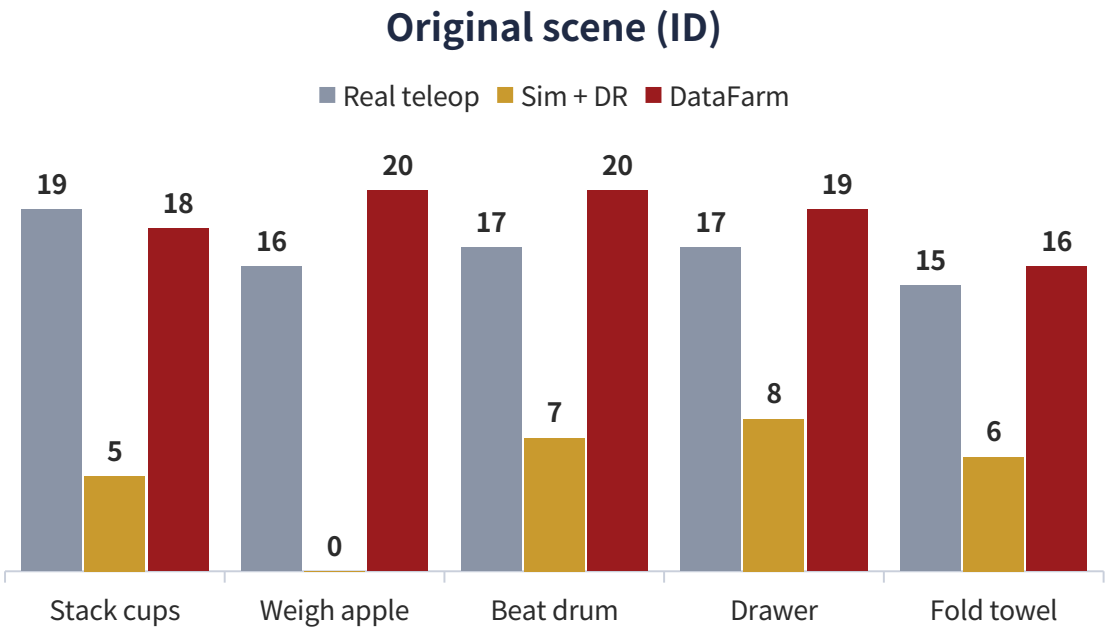
OOD: teleop nearly halves; the farm leads on every task

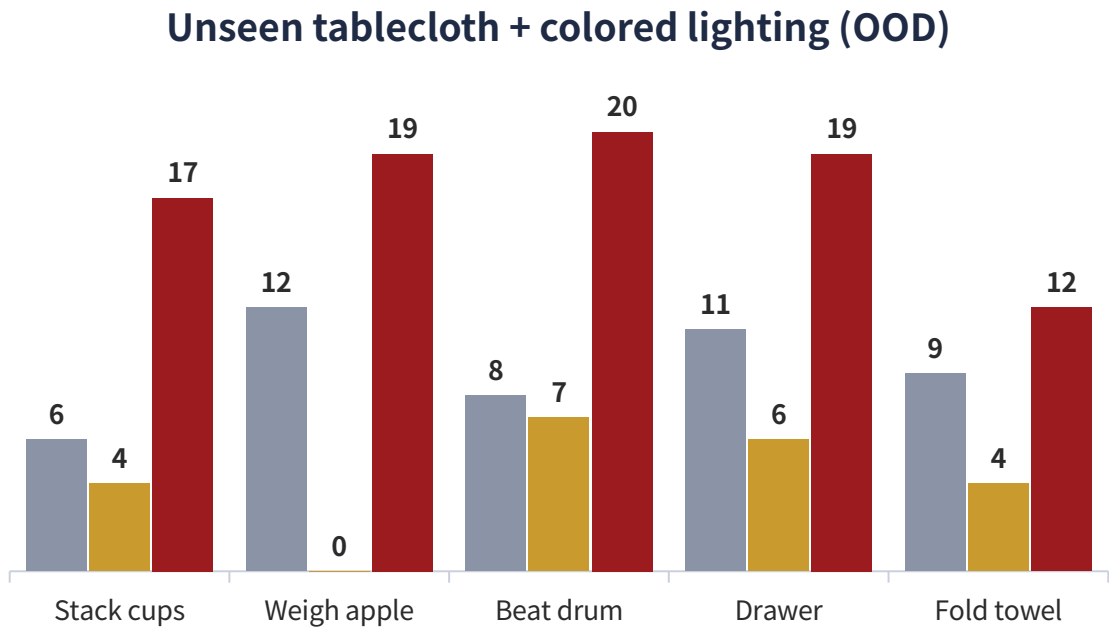
26 / 21

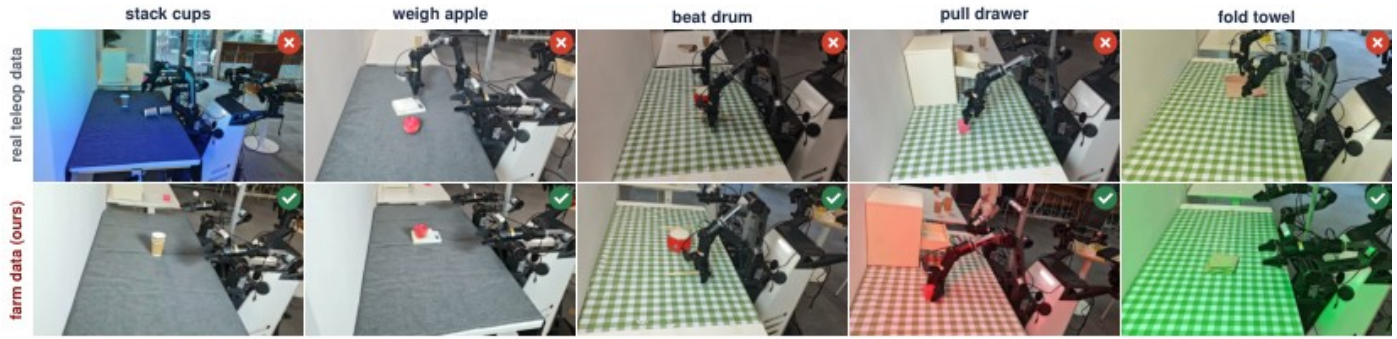
Sim + DR: actions identical to the farm, only sim-rendered → the gap is the contribution of "rendering with real images"

Source: Fig. 4(a), § 6.1 (successes / 100 trials)

Per task: under unseen looks, the farm policy leads on all 5 tasks

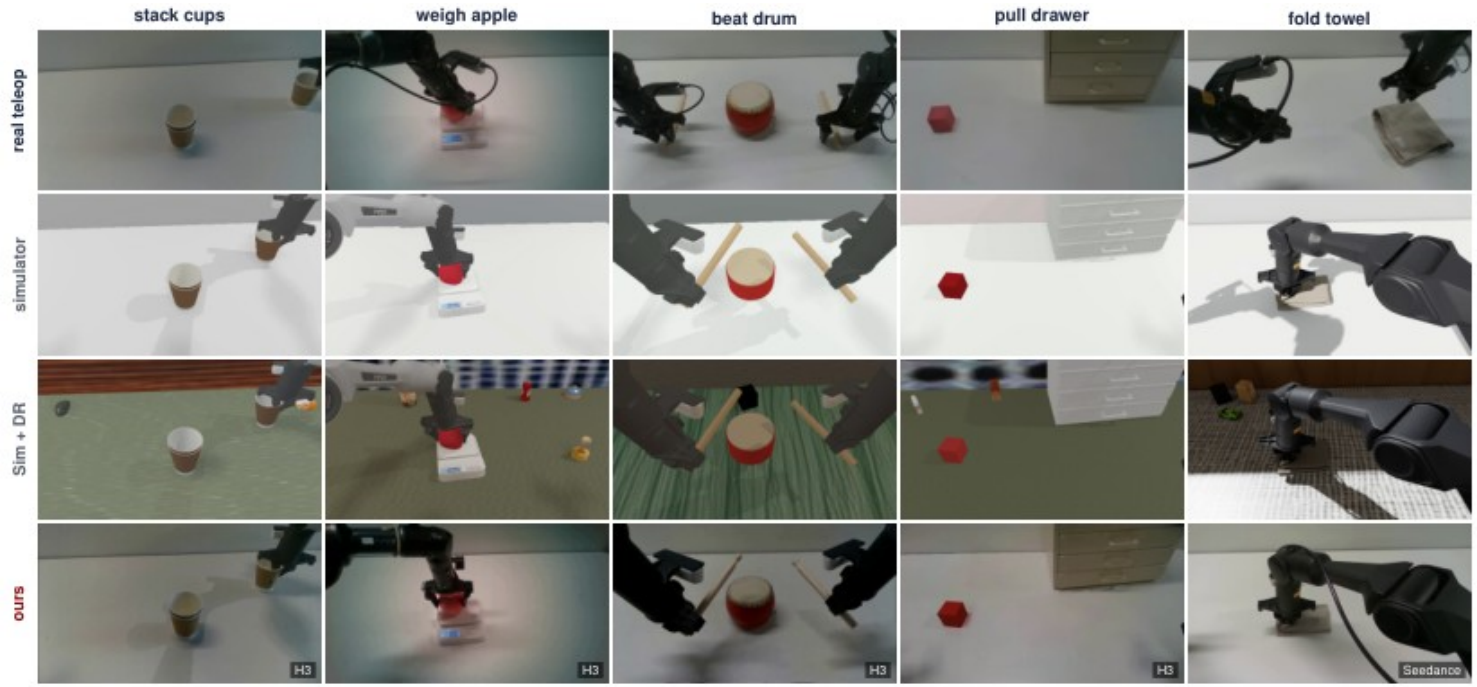




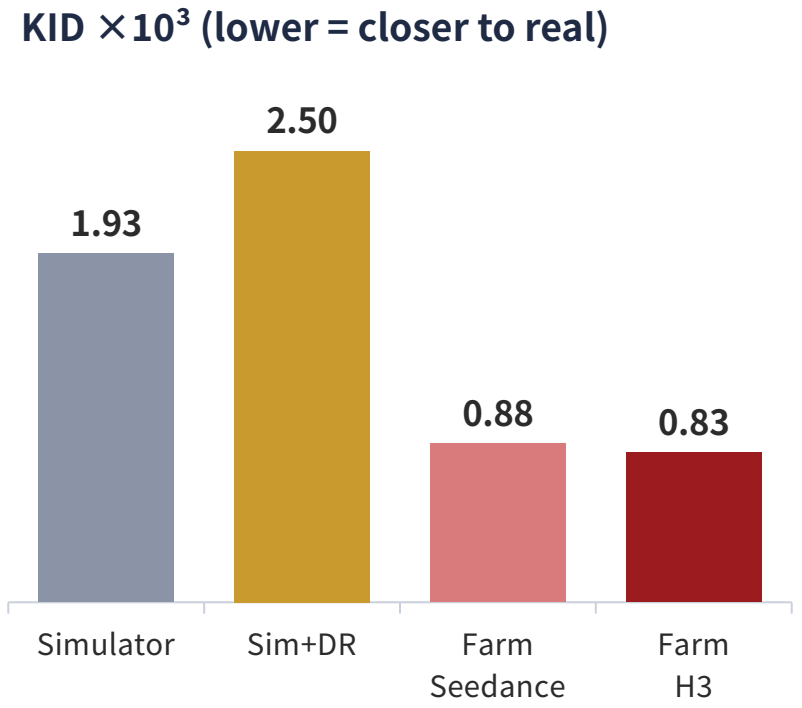


Below: OOD examples; top row teleop policy fails (✗), bottom row farm policy succeeds (✓). Bar labels: successes out of 20, read from Fig. 4(a).

Realism: farm frames are under half as far from real as sim



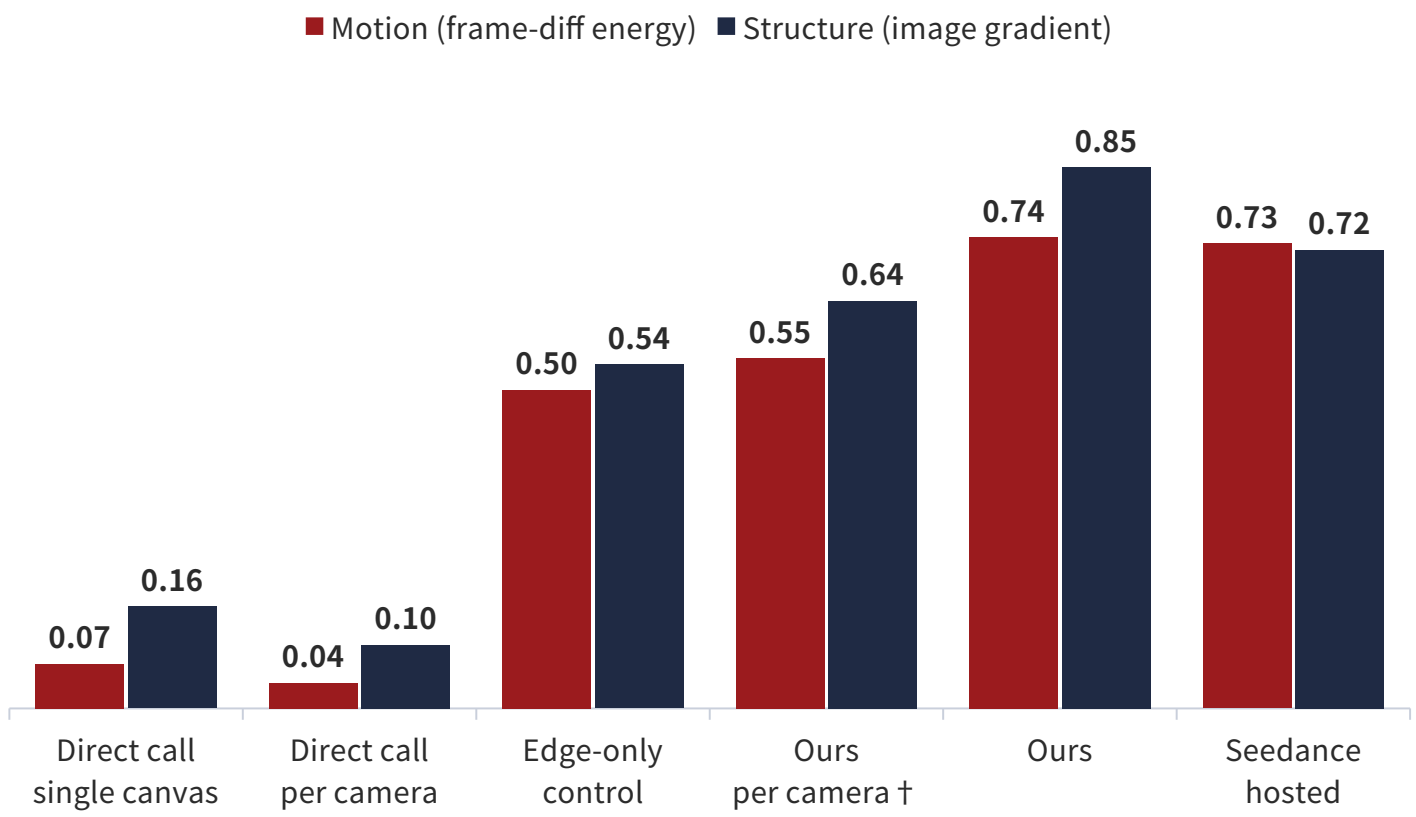
Head view at the same task stage: real teleop / simulator / Sim + DR / ours. Real and sim frames come from different episodes, with different object layouts.



- DINOv2 features, head view, mean over cup / apple / drawer tasks
- Farm is closer to real than sim on all 5 tasks
- Domain randomization is even farther: random textures look unlike real scenes

Joint multi-view rendering lifts motion correlation with sim to 0.74

Correlation with sim video (higher is better; mean of three clips)



† Cup-stacking clip only; ours scores 0.75 / 0.87 on it. Direct call = MiniMax-H3 with sim video as motion reference, real reference frames and contract prompts.

30.2 → 2.9

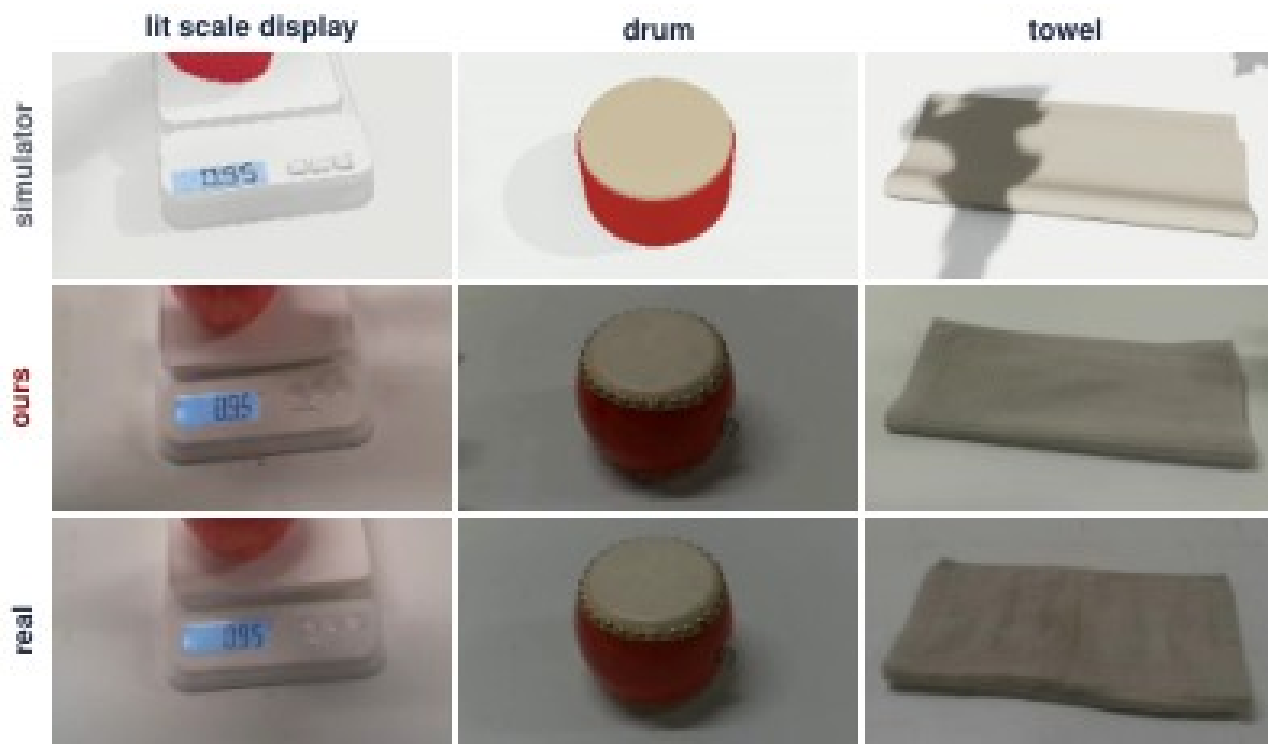
Jumps per 1k frames: edge control only → plus references, prompts and chunk stitching

Optical-flow check (Fig. 6, 5-task mean)

Source	Flow error	Consist. px
H3 render	4.4 px	65%
Sim + DR (identical motion)	3.4 px	73%
H3 shifted 0.5 s (control)	8.8 px	41%

H3 nears Sim + DR (identical motion) and far beats the shifted control, so rendering preserves sim motion; drumming with a fast, thin stick has the largest error.

Video model fixes sim-hard looks: lit screen, painted drum, towel



Objects	Simulator	Farm render
Scale display	Flat texture	Lit screen showing sim reading
Drum	Red cylinder	Lacquered shell + brass studs
Towel	Smooth particle cloth	Real towel weave

What Real2Sim2Real needs

Model glowing displays, scan the drum, build textures that deform with cloth, and recapture for every new scene.

What the farm needs

Geometry and contact needed for physics + a $4.1–6.2 per-task render contract; motion and object state still come from sim.

Source: Fig. 9, Table 3, § 6.5

Cost: ~$26 one-time per task, then $0.80 per episode

$20.92
One-time adaptation (5-task mean)
grounding → expert admission

$5.13
Render contract (mean)

$0.80
Marginal cost per episode m
Sim + render + re-render + review

$107
First 3-view hour (w/ one-time)
Then $81 per hour

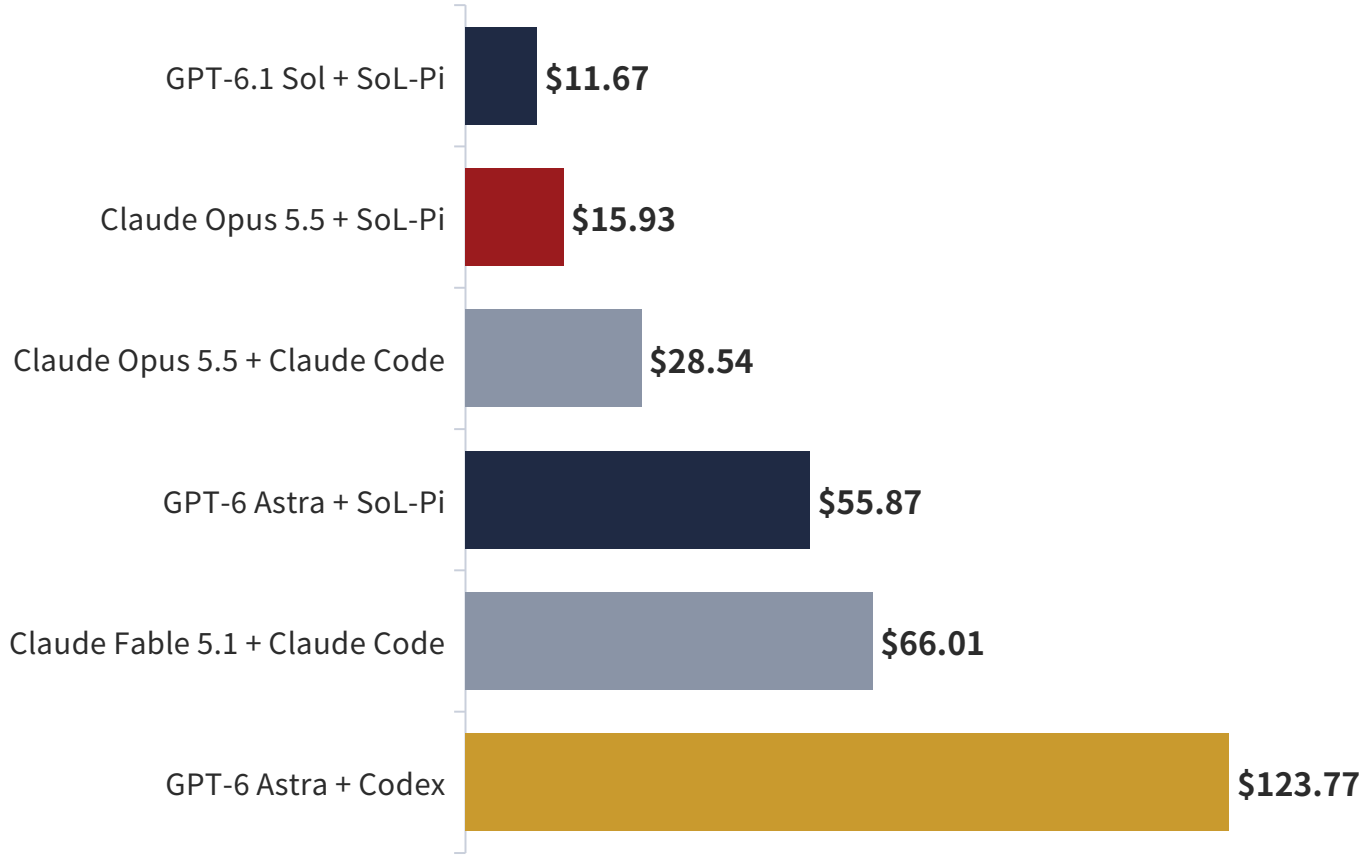
Task	One-time	Render contract	Per-ep. m	Avg/ep. N=100	Avg/ep. N=1000	Each extra hour	First hour
Stack cups	18.22	6.11	0.93	1.17	0.96	77	101
Weigh apple	10.33	4.72	0.89	1.04	0.91	102	117
Beat drum	19.25	4.11	0.26	0.49	0.28	83	106
Drawer	19.40	4.49	0.52	0.76	0.54	55	79
Fold towel	37.38	6.23	1.41	1.84	1.45	88	132
Mean	**20.92**	**5.13**	**0.80**	**1.06**	**0.83**	**81**	**107**
Teleop • N. America (labor only)	–	–	2.86	2.86	2.86	290	290
Teleop • China (quote)	–	–	1.16	1.16	1.16	117	117

- Units: USD. Break-even: after 10–37 episodes per task vs. North American labor, and after 41–220 vs. Chinese quotes.
- Assumes self-hosted H3 (B200, $3/GPU • h); at Seedance list prices, rendering one episode costs $12–63. NA figures are estimates; China is a single all-inclusive quote, so bases differ.

Source: Table 3, Table 4, Fig. 5(a), § 6.2, App. D

Every agent works, but cheap does not mean reliable

One-time cost (USD; mean of cup / apple / drum tasks)



Colors: navy SoL-Pi, gray Claude Code, gold Codex; red marks the combo used for later costs. Last two rows are earlier runs under a different protocol.

All 6 combos pass admission

- Mean $12–124 per task
- Differences mainly track token prices

Cheapest combo has hidden risks

- Cups and drum needed extra human instructions
- 4 of 5 admitted drum clips exceed the 3 rad/s joint speed limit, unchecked by its own review

Opus 5.5 + SoL-Pi chosen

- No human intervention; its review rejected and fixed the same overspeed defect

Source: Fig. 5(b), Table 8, App. D (single run)

Caveats: evidence scale, comparison fairness, cost assumptions

Small evaluation

20 trials per task per condition; per-task ID gaps of 0–4, so 93 vs 84 is better read as parity. Only one bimanual robot and 5 tabletop tasks.

Authors acknowledge / reviewer adds

OOD comparison favors farm

Farm training data include text-specified appearance changes; teleop has only the original scene, unaugmented. Missing: a "teleop + generative augmentation" or mixed-data baseline.

Reviewer's view

Simulator-dependent

Mass and friction are only randomized, not identified, possibly inadequate for fine-dynamics tasks; skill probes exist only for cloth so far.

Authors acknowledge

Costs assume self-hosted render

At Seedance list prices ($12–63 per episode) the advantage vanishes; NA is an LLM-assisted labor-only estimate, China a single all-inclusive quote.

Authors disclose

Video checks fallible

Automated checks miss things; a cheap combo admitted clips exceeding joint speed limits; coverage depends on check design.

Authors acknowledge

Motion fidelity still lags

Flow consistency: H3 65% vs Sim+DR 73%; fast thin objects (drumstick) err most; towel folding is weakest OOD (~12/20).

Reviewer adds

Three Takeaways

01

Agents can take over the human bottleneck of "building tasks"

One calibration video + one request → scene, cameras, expert, hidden checker, render contract; fully automatic and task-agnostic.

02

Actions from simulation, appearance from real footage

On par with equal-size teleop on the real robot (93 vs 84), more robust under unseen looks (87 vs 46), far above Sim + DR on the same trajectories (26 / 21).

03

Trust comes from separation of powers, not from trusting the agent

Hidden checkers, simulator certificates, mutation-calibrated verifiers, independent review and a full ledger; first-hour data costs $107, below $117 (China quote) / $290 (NA labor).

Still to verify: more robots and task families, fair comparison with "teleop + generative augmentation", and tasks relying on fine dynamics.